Fraudulent Claim Detection

Problem Statement

Global Insure faces significant financial risks due to fraudulent insurance claims.
Currently, fraud detection is manual, leading to delays and increased payouts.

The business goal of this assignment is to develop a machine learning-based fraud detection system that classifies claims as fraudulent or legitimate**,** enabling early intervention and reducing financial losses losses and optimise the overall claims handling process.

Methodology

1. **Data Understanding**

   Analyzed claim data containing customer information, policy details, claim amounts, and incident descriptions and about the fraud reported.

2. **Data Cleaning**

   a) Address the missing values- Removed irrelevant columns (e.g., _c39),
   b) Handled missing values - like police_report_available.
   c) Identify and handle redundant values
   d) Identify and drop rows where features have illogical or invalid values
   e) Fixed the data type- datetime data type

3. **Train-Validation Split**

   Correctly defining the feature and target variables.Split the data into training and validation into 70% train data and 30% validation datasets to prevent overfitting and ensure proper model evaluation.

4. **Exploratory Data Analysis (EDA)**

   a) Performed univariate analysis by visualising the distribution of all numerical columns
   b) Performed correlation analysis by visualising a heatmap of the correlation matrix
   c) To examine class imbalance - Visualised class distribution of the target variable with help of box plot
   d) Performed bivariate analysis by visualising the relationship between categorical columns and the target variable

5. **Feature Engineering**

a) Use the **RandomOverSampler** technique to balance the data and handle class imbalance.

b) Constant- Create new features based on your understanding for both training and validation data.
c) Drop- Handled redundant columns (eliminate any columns from which the necessary features have already been extracted in the preceding step).
d) Dummy variable - Transform categorical variables into numerical representations using dummy variables on both training and test data set.
e) **StandardScaler** from scikit-learn was used to numerical columns . It helps the logistic regression model converge faster and assign balanced weights to features.

6. **Model Building**

Two machine learning models were built: **Logistic Regression** and **Random Forest**, following a structured process to ensure optimal performance:

a) Feature Selection:
Applied **Recursive Feature Elimination with Cross-Validation (RFECV)** to select key features.
b) Model Building:

- -For Logistic Regression, checked multicollinearity using **p-values** and **VIF**.
- ained both models on the training data and optimized hyperparameters through validation.

c) Performance Evaluation:

- Used appropriate metrics to evaluate predictions.
- Created a cutoff optimization plot to balance **recall, F1-score, sensitivity, and specificity** at different thresholds.

7. **Prediction and Model Evaluation**
Evaluated models using **Precision, Recall, and F1-Score**, with focus on fraud detection accuracy.

Techniques Used

a. Data Preprocessing:
-Removal of noisy columns, encoding of categorical features, handling missing data.
b. Oversampling:
-Applied Random OverSampling (ROS) to balance the dataset.
c. Modeling Algorithms:
-Random Forest Classifier (best-performing)
-Logistic Regression
d. Evaluation Metrics:
-Recall to minimize missed fraud cases.
-Precision and F1-Score for balanced assessment

Assumptions

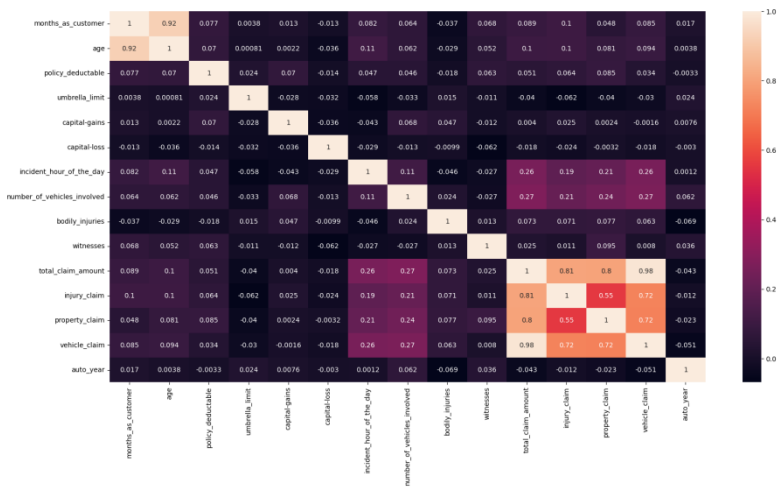
- Redundant columns were droped
- Dummies were created
- optimal_cutoff is 0.48
- Fraud rates in real-world deployment may be lower than in the dataset; hence, precision might drop.
- Model assumes availability of features at claim submission time for real-time predictions.
- The current approach focuses only on structured data, not text or images.

Visualisation

a) Plot distribution of numerical columns

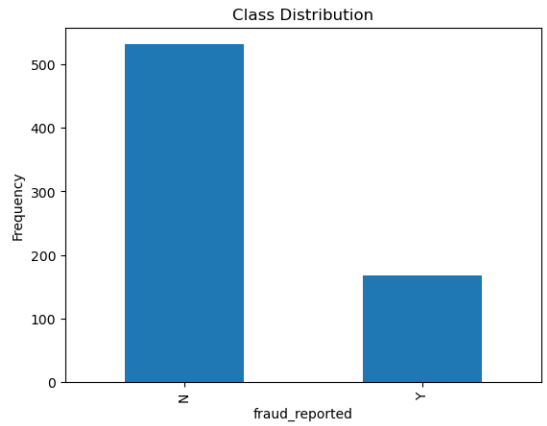
- ❖ There are high correlations between financial variables, requiring careful feature selection to avoid multicollinearity.
- ❖ Claim-related features show skewed distributions.

b) correlation matrix for numerical columns



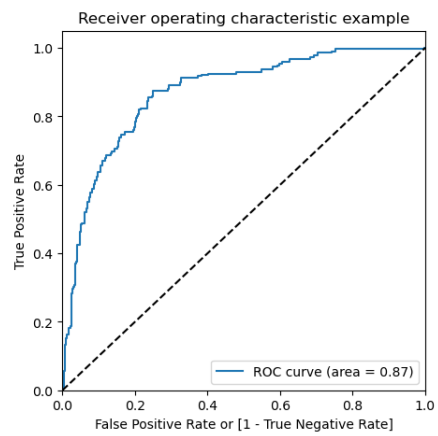
- ❖ **months_as_customer** and **age** show a **very strong correlation** of **0.92**.
  This indicates that older customers tend to have longer relationships with the company.
- ❖ Most other variables show low correlations (close to 0), suggesting they contribute independent information, which is good for predictive modeling

c) The bar chart shows the distribution of the target variable fraud_reported



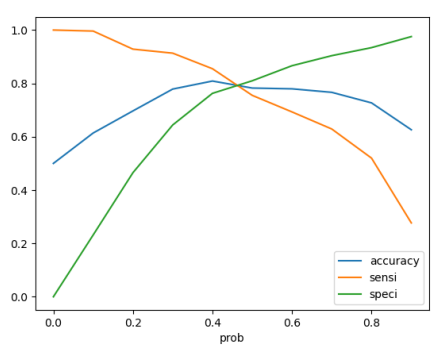
- The dataset is highly imbalanced**,** with fraud cases being the minority**.**
- Fraudulent claims are about 24% of the total data, while non-fraudulent claims dominate

d) ROC



- ROC curve (area = 0.87). It indicates that the Area Under the Curve (AUC) for this specific ROC curve is 0.87.
- Blue curve shows trade of between TPR and FPR. Roc curve is top left corner so its better model.

e) Plot accuracy, sensitivity, and specificity at different values of probability cutoffs

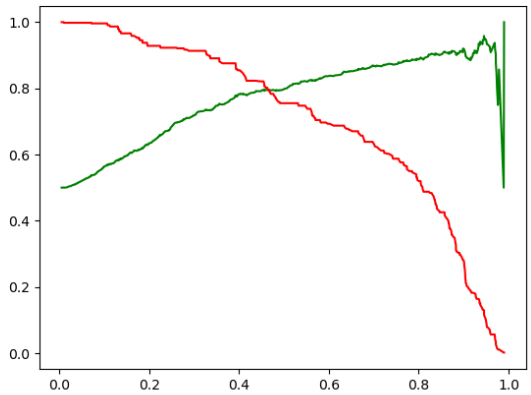


- There's a clear inverse relationship between sensitivity and specificity. As one

increases, the other tends to decrease.

- ❖ In this plot, the "sensi" and "speci" lines intersect around a `prob` of approximately 0.4, which also seems to be near where "accuracy" peaks.

f) Precision-Recall curve



- ❖ The point where these two lines intersect or are both high (around the 0.4-0.5 mark on the x-axis in this specific plot) often represents a good balance.

Model Results

MODEL	PRECISION	RECALL(Fraud)	F1 SCORE
Logistic Regression	0.54	0.68	0.60
Random Forest	0.57	0.65	0.61

- Random Forest achieves better precision and F1-score, offering a more balanced performance between fraud detection and false alarms.
- The fraud detection context, where missing a fraud is more costly than investigating a few false alarms, **Random Forest** is recommended for deployment due to its overall balanced performance

Key Insights

1. **Fraud is Rare but Critical**
2. Only **24% of claims are fraudulent**, highlighting the need for focused detection to reduce financial losses.
3. **Financial Variables Drive Fraud Detection**
4. **Claim amounts and components** are key fraud indicators but require careful handling due to multicollinearity.
5. **Random Forest Performs Best**

6. **Random Forest** achieves better **precision (0.59)** and **F1-score (0.59)**, making it the **preferred model** over Logistic Regression.
7. **Cutoff Optimization Matters**
8. An **optimal cutoff of 0.48** balances sensitivity and specificity, improving fraud capture without excessive false positives.
9. **Robust Model Validation**
10. **ROC AUC of 0.87** confirms strong model performance.

Conclusion

The project successfully developed a machine learning-based fraud detection system, providing Global Insure with a **proactive, data-driven approach** to identifying fraudulent claims before approval.